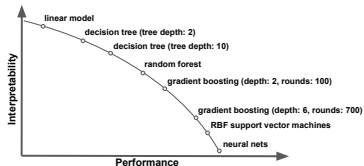
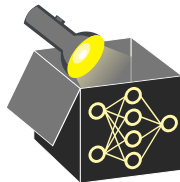


Interpretable Machine Learning

Intro to IML

Introduction, Motivation, and History

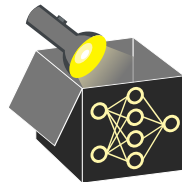


Learning goals

- Why interpretability?
- Developments until now?
- Use cases for interpretability

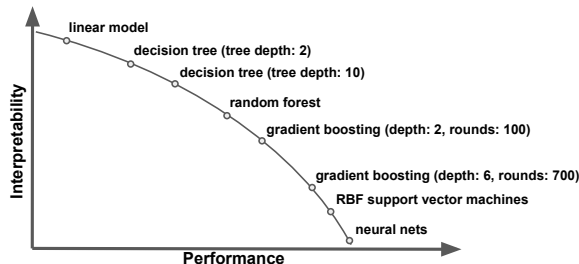
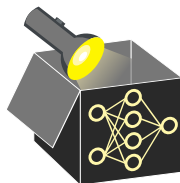
WHY INTERPRETABILITY?

- ML aids decision-making in applications through strong predictive performance
- ML models are black boxes, e.g., XGBoost, RBF SVM, or DNNs
 $\rightsquigarrow$ too complex to be understood by humans
- Some applications are “learn to understand”, not “learn to predict”



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→ too complex to be understood by humans
 - Some applications are “learn to understand”, not “learn to predict”
 - When deploying ML models, lack of explanations
 - ❶ reduces trust
 - ❷ creates barriers to adoption
- Many disciplines with required trust rely on traditional models, e.g., linear models, despite lower predictive performance



INTERPRETABILITY IN HIGH-STAKES DECISIONS

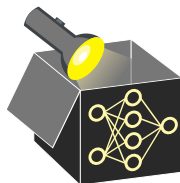
Examples of critical areas where ML-based decisions can affect human life

- Credit scoring and insurance

applications

► SOCIETYOFACTUARIES2021

- Reasons for not granting a loan
- Fraud detection in insurance claims



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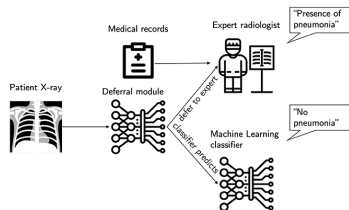
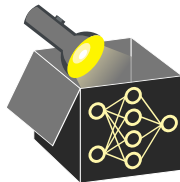
applications ▶ SOCIETYOFACUARIIES2021

- Reasons for not granting a loan
- Fraud detection in insurance claims

- Medical applications

- Identification of diseases
- Recommendations of treatments

- ...



▶ Click for source

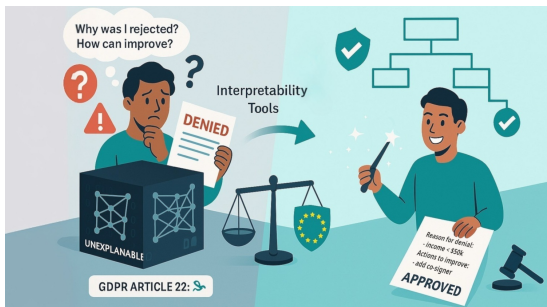
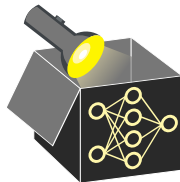
NEED FOR INTERPRETABILITY

Need for interpretability becomes increasingly important from a legal perspective

- General Data Protection Regulation (GDPR) requires for some applications that models have to be explainable ▶ GOODMANFLAXMAN2017

↪ *EU Regulations on Algorithmic Decision-Making and a “Right to Explanation”*

- *Ethics guidelines for trustworthy AI* ▶ EUROPEANCOMMISSION2019
- *The EU Artificial Intelligence Act* ▶ EUROPEANCOMMISSION2024



BRIEF HISTORY OF INTERPRETABILITY

- 18th/19th century:
Linear models (Gauss, Legendre, Quetelet)
- Mid-20th century:
Emergence of sensitivity analysis
- Mid- to late 20th century:
Rule-based ML, incl. decision rules and trees
- Early 2000s:
Interpretation tools for ensembles:
 - variable importance for random forests
 - partial dependence for gradient boosting
- Early 2010s:
Explainability methods for deep learning
- Late 2010s:
IML / XAI as a distinct research area



► GAUSS1828

► WIKIPEDIA GAUSS

